ICRA 2018 DJI RoboMaster AI Challenge

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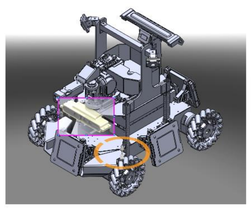
* : Those team members use this project as their independent project for ME 530.707 Robot Systems Programming taught by Prof. Louis Whitcomb at Johns Hopkins University.

Abstract

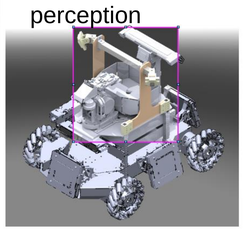
Artificial Intelligence is an emerging field with exciting development in the recent years. As one of the best attempts to build a standard platform for AI algorithm developing, DJI RoboMaster AI Challenge is a great opportunity for us to develop exciting algorithms addressing real world issue.In this article, we will report the progress that we have made thus far in the preparation for theRoboMaster AI Challenge. The report is divided into Hardware and Software sections, and bothhave subsections that further elaborate the technical details

Hardware

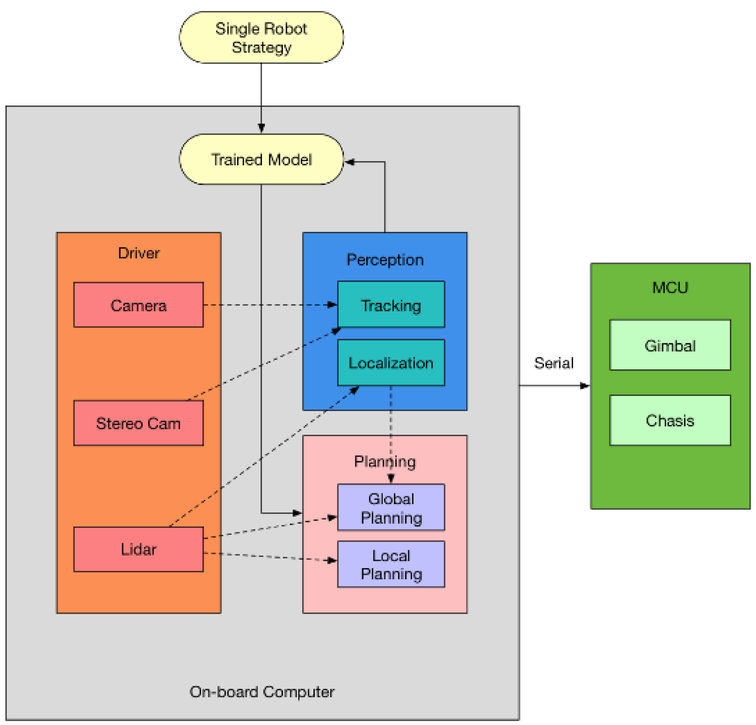


RGBD camera

two more cameras are implemented to enhance the perception

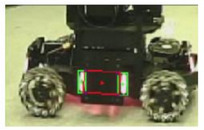


Software Architecture



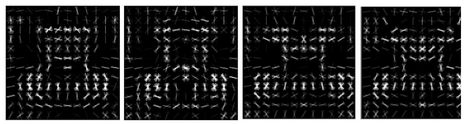
Enemy Perception

(a) Extracted color areas.



(b) Color detection results

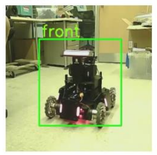
The perception module detects and tracks enemy robots and armors. The top figures show the detection of the LED lights on the side of armor packs. The figures on the right and below show the effort in detecting and tracking the robot, as well as trying to identify its orientation.



(a) Front HOG descriptor (b) Back HOG descriptor (c) Left HOG descriptor (d) Right HOG descrip

(a) Front correctly detected.

(b) Right correctly detected.



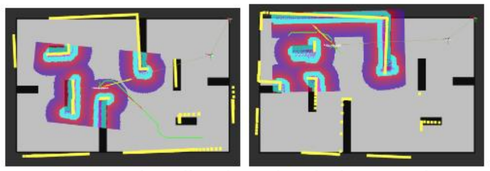
(c) Back wrongly detected as back.

(d) When detection fails, KCF tracking fills in.

(e) Vehicle occluded.

(f) Tracker recovered.

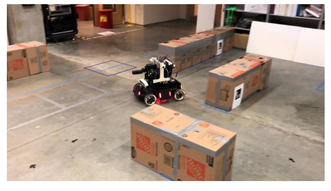
Localization and Navigation

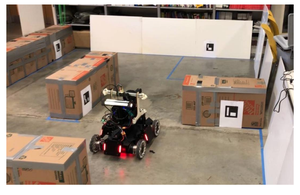


AMCL localization simulation results.

Localization: At the implementation level, the AMCL package from ROS represents the probability distribution using a particle filter. Now as the robot moves forward, we generate new samples that predict the robots position after the motion command. Sensor Readings from Lidar are incorporated by re-weighting these samples and normalizing the weights. The package also requires a predefined map of the environment against which to compare observed sensor values, where we can get the map by gmapping.

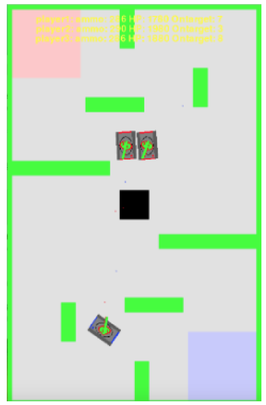
Navigation: When navigating the robot through a mapped environment, the global trajectory is calculated by A* algorithm. Meanwhile, Time-Elastic Band is used to optimize local trajectory based on execution time, distance and heading difference with respect to the goal, separation from the obstacles and the kinodynamic constraints of the robot itself.



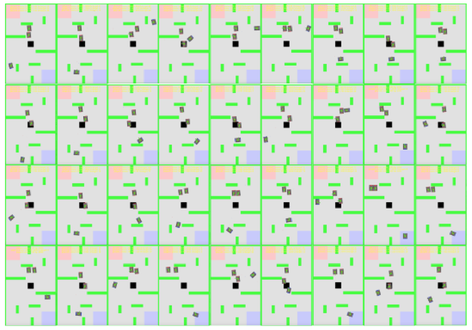


Figures showing robots navigating the mock arena.

Strategy and Decision Making



Inspired by recent research and breakthrough such as AlphaGO and OpenAI, we decided to reinforcement learning to teach our robot the best strategy. A part of that effort was devoted to make the game shown on the left, in which the player has control over the movement of the blue robot. The canon aims and fires automatically like the actual robot. At this moment, we are still tuning for better performance. Below is a figure showing learning in progress.



Besides the attempt to apply reinforcement learning, we are also developing a Finite State Machine(FSM) model which dominates the robot behavior based on predefined strategies.

Next Steps

As one of the 22 qualified out of 67 teams globally, we are going to participate in the challenge at this year's ICRA in Brisbane, Australia May 21st - 24th. Therefore, we will put most of our effort in building the remaining of the system. So far, we still have work left in Enemy Perception, Strategy and Decision Making, and Architecture. More specifically, we need to: 1. Increase the robustness of the perception algorithms; 2. Finish training and transfer learning; 3. Finish FSM design and implement; 4. Integrate the system into ROS.

Since the beginning of this project, we received positive feedback from faculty members as well as student community. Considering the lack of a competitive student robot competition team, we are planning to form a student organization based on LCSR. The purpose of this organization is to provide students interested in Robotics more hand on experience by participating in various robot competition.

Acknowledgments

The project is supported by Prof. Charbel Rizk, Electrical and Computer Engineering department at Johns Hopkins University, and Prof. Louis Whitcomb, Laboratory of Computational Sensing and Robotics, Johns Hopkins University. We thank for their sponsorship and their lab space offering.